To the Editors,
International Journal of Heat and Mass Transfer

Dear Editors,

Please consider our manuscript, "Field accuracy is not heat accuracy: wall-flux-constrained surrogates for non-Boussinesq convection", for publication in the *International Journal of Heat and Mass Transfer* as a full-length article.

Surrogates for thermal convection are usually trained on paired Boussinesq and compressible solutions. Public paired data sets fix the relative temperature contrast, so a network trained on them memorises one correction field and cannot rescale it at another contrast. The manuscript asks whether the metric used to select these surrogates is the right one. On 200 held-out classical-cavity samples, a model with a 0.09% median temperature-field error misses the both-wall integral Nusselt number by 2.0% on average (1.4–2.9% across seeds); in 42% of the samples on average the field error is below 1% and the Nu error above 2%. On a 22-sample random-wall replication at $\varepsilon \approx 1.2$, the corresponding field and Nu errors are 0.47% and 4.04%, with 66% on average in the same region. A wall-flux penalty in the training loss reduces the both-wall error under the training six-point stencil to 0.01% on the classical cavity and 0.58% in the replication, and the local wall-flux error to 0.08% and 1.14%; the median field error moves from 0.09% to 0.10% and from 0.47% to 0.49%.

The underlying correction factorises as $D = \varepsilon P(X) + O(\varepsilon^2)$ with $\varepsilon = \Delta T/T_0$. Across 627 paired solutions, the velocity correction follows $R = A\varepsilon^m$ with exponents indistinguishable from one; a quadratic refinement gives positive curvature $\kappa = 0.70$ and 0.83 (point estimates), and the prefactor is a functional of the flow. Training a network on $P = D/\varepsilon$ and multiplying by ε outside makes a public single-contrast data set reusable within its boundary-condition family, reducing the channel-averaged extrapolation error from 0.131 to 0.085. A plain model trained at the same contrast and rescaled by $\varepsilon/\varepsilon_0$ performs within 0.2% of the factorized target, so applying ε outside the learned correction, rather than the residual parameterisation, is what makes the data reusable.

The study is within the journal's scope: it quantifies the departure from the Oberbeck–Boussinesq approximation, tests it on independently generated public data, and provides a surrogate design that preserves wall heat transfer. The main sweep is two-dimensional and steady; a 33-sample three-dimensional check supports the exponent, and the remaining scope is stated explicitly. The 256^2 Boussinesq solution is within 0.37% of the Le Quéré benchmark at $Ra = 10^6$. The work is original, is not under consideration elsewhere, and all authors have approved the submission. The authors declare no competing interests. The public data are cited, and our generated data, solver configuration and reproduction scripts are available at the repository listed in the manuscript.

Yours sincerely,
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